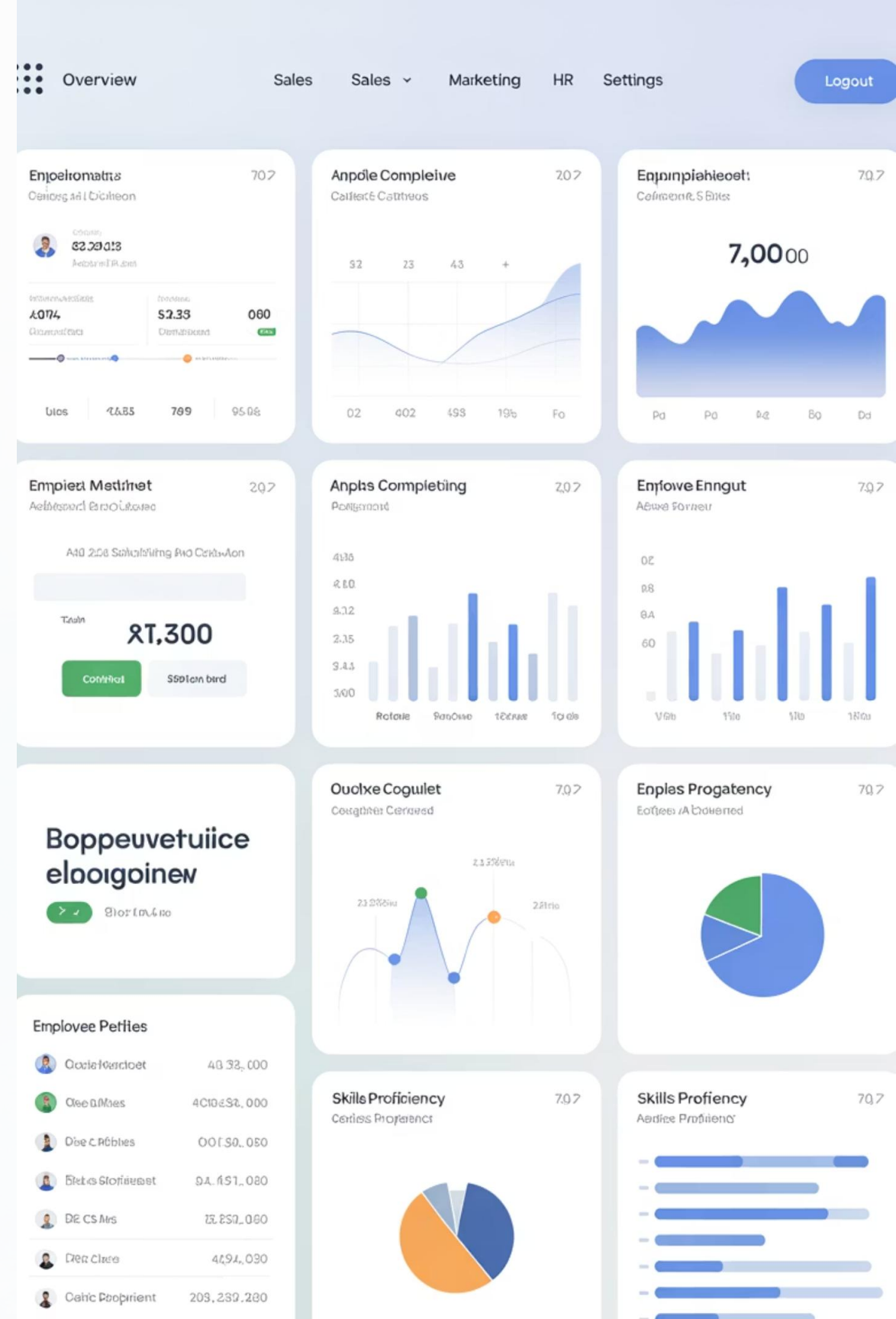


Employee Performance Prediction

Data Analysis to Improve Organizational Performance



Agenda

Introduction to Performance Prediction

Importance and Evolution of Performance Management

Performance Measures

Key Indicators and Measurement Methods

Analytical Methods

Statistical Techniques for Performance Prediction

Case Studies

Real Examples of Predictive Analysis

Ethical Considerations

Ethical Aspects in Performance Data Analysis

Introduction to Performance Prediction

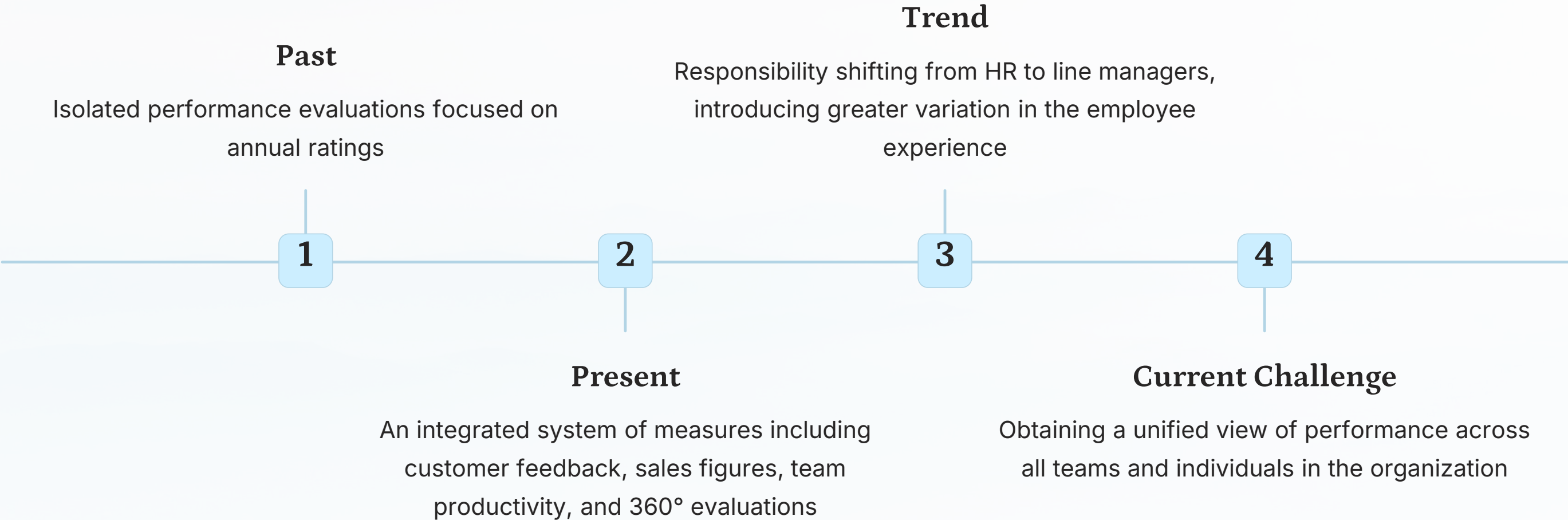
In response to volatile markets and increased economic uncertainty, organizations have placed greater emphasis on individual, team, and organizational performance.

The analytical capability of HR to predict future performance is considered the “holy grail” of HR analytics.

Identifying, predicting, understanding, and potentially influencing high levels of performance is critical to improving organizational performance and guiding people strategy.



Evolution of Performance Management

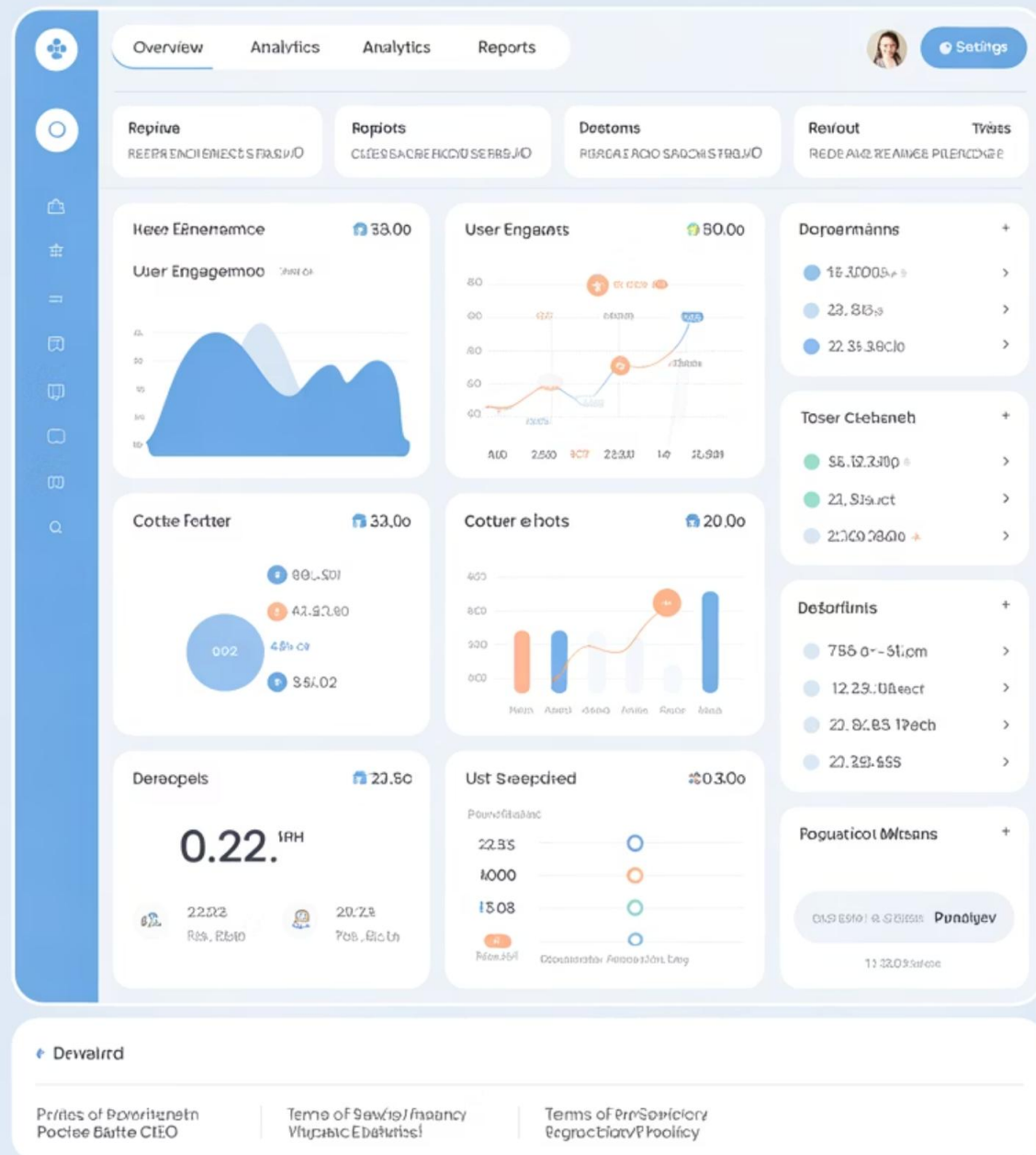


Decentralization makes it difficult to achieve a consistent and comprehensive view of performance across all indicators.

What Can We Measure to Indicate Performance?

One of the main challenges in conducting performance analysis with HR data is deciding which type of performance to predict.

It is important not only to measure performance, but also to consider which metrics could serve as predictors of performance.



Performance Indicators

Performance Evaluations

Annual or biannual performance ratings

Customer Satisfaction

Customer loyalty and likelihood of repeat business

Operational Metrics

Scan rates, call duration, sales

HR Metrics

Sick leave absence, employee turnover, engagement scores

With the shift from manufacturing industries to the service sector, customer satisfaction has become a primary measure of performance.

Analytical Methods



Multiple Linear Regression

Analyzes multiple independent variables to predict a dependent variable by building a “best-fit” model



Stepwise Regression

Performs iterative steps to identify the most parsimonious model that best fits the data



Tracking Over Time

Tracks values over time to identify patterns of behavior



Dummy Variables

Manipulates data to indicate the absence or presence of specific characteristics

Case Study: Customer Loyalty Prediction

Analysis of a financial services organization using customer satisfaction survey data.

Independent Variables:

- **Understanding of customer needs:** *The sales person understanding your needs*
- **Trust in the salesperson:** *The sales person seems confident*
- **Recommendation of the salesperson:** *The sales person has a recommendation*
- **Salesperson's knowledge:** *The sales person is knowledgeable*
- **Salesperson's gender**



Key Findings: Understanding customer needs and giving a recommendation are the main drivers of customer loyalty

Case Study: Individual Performance Prediction

Analysis of a manufacturing organization using performance appraisal data, sick leave records, and employee attitude surveys.

Independent Variables:

- **Perceived organizational support**
- **Perceptions of distributive justice**
- **Perceptions of procedural justice**
- **Job satisfaction**
- **Person–organization fit**
- **Job strain**
- **Sick days**



Case Study: Change in Performance Over Time

Using multiple stepwise regression to model changes in performance over time, controlling for prior levels of performance.

Regression Model:

Performance ratings 2015 = 0.467 + 0.377 (performance ratings 2014) + 0.173 (procedural justice) + 0.175 (job strain) – 0.016 (sick days)



Case Study: Performance Patterns by Demographic Data

Analysis of a large multinational manufacturing organization using data from 9,450 employees across four countries.

Independent Variables:

- **Benchmark salary**
- **Gender**
- **Age category**
- **Tenure category**
- **Country (U.S., United Kingdom, Netherlands, Russia)**



Case Study: Supermarket Scan Rates

Exploration of scan rate patterns at supermarket checkouts based on employees' demographic characteristics.

Dependent Variable: Average number of items scanned per minute (per employee).

The target in this particular supermarket was 22 items per minute.



Many supermarkets set scan rate targets for employees working at the checkout.

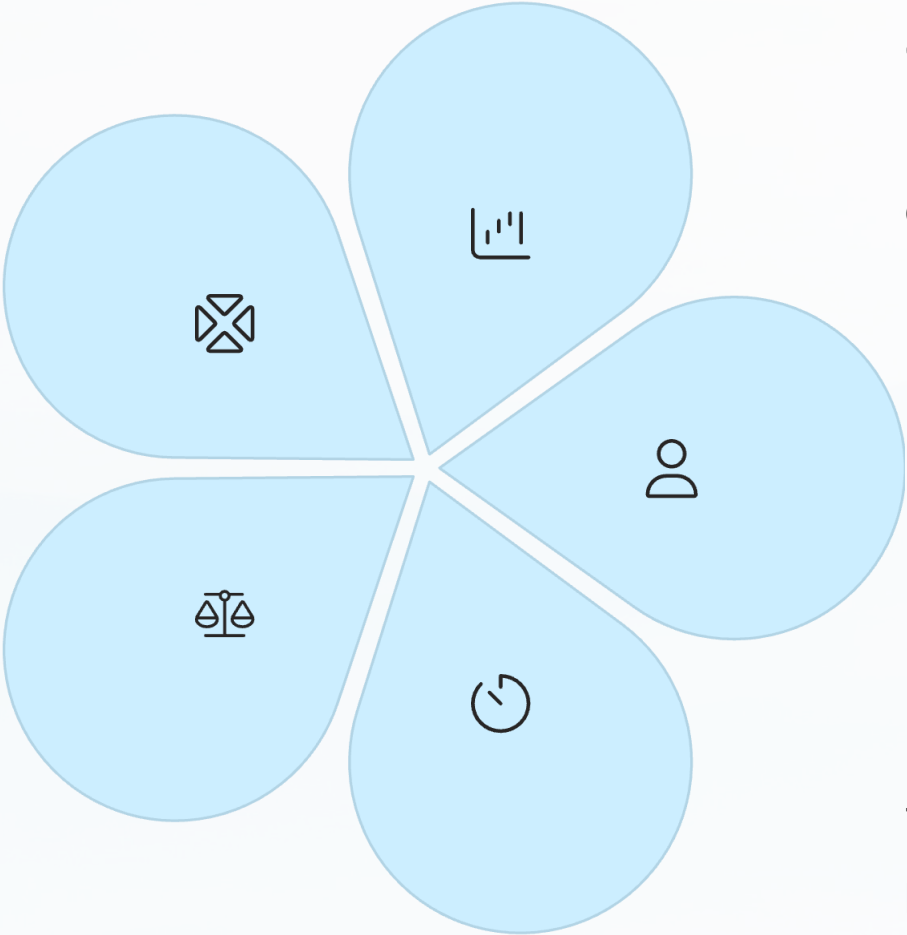
Key Conclusions

Multiple Indicators

Use a variety of performance indicators to obtain a comprehensive view

Balanced Approach

Apply a balanced scorecard approach to avoid reliance on single metrics



Contextual Analysis

Interpret results considering the specific organizational context

Ethics First

Consider the ethical implications of findings and recommendations

Longitudinal Analysis

Track performance over time to identify patterns and trends

HR analytics can provide valuable insights into the factors that predict performance, but these findings must be interpreted and applied with care.